The Epoch-Level Illusion: An Honest Subject-Wise Benchmark of EEG Epilepsy Seizure Detection within a Responsible, Agentic, Retrieval-Augmented Deployment Framework (RGAIG)

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***Abstract*—Background. Automated EEG seizure detection is routinely reported above 95% accuracy, but such figures are almost always obtained under *epoch-level* cross-validation, which leaks subject-specific information across train/test folds and overstates deployable performance. Methods. Holding feature representation and classifier fixed, we vary *only* the validation protocol. A 20-dimensional feature pipeline (per-channel $\delta\theta\alpha\beta\gamma$ relative band power, three Hjorth descriptors, line-length, RMS; mean+std across channels over 8 s epochs) feeds a class-balanced Random Forest, evaluated under (i) 5-fold stratified CV on Bonn/UCI (11,500 segments) and (ii) strict Leave-One-Subject-Out (LOSO) on CHB-MIT (24 subjects). We add hyperparameter, statistical, sensitivity, ablation, and failure-mode analyses, and situate the detector in a five-layer governed deployment stack (RGAIG). Results. Epoch-level evaluation yields 96.99% accuracy (AUC 0.996, sensitivity 88.4%). The identical pipeline under LOSO yields 90.0% accuracy and AUC 0.846, but mean sensitivity collapses to 35.1% with a per-subject range of 0–89.7%. Specificity (96.9%) and NPV (92.3%) remain high. Conclusion. The 53-point sensitivity gap is a measurement artefact of epoch-level validation, not a modelling deficiency. Subject-wise sensitivity, not epoch-level accuracy, is the only honest figure of merit. We release all code and per-subject results.**

Keywords: EEG, epilepsy, seizure detection, Leave-One-Subject-Out, data leakage, responsible AI, retrieval-augmented generation, Model Context Protocol, AIOps, reproducibility.

I. Introduction

Epilepsy affects ∼50 million people worldwide and continuous EEG remains the diagnostic gold standard [1]. Recent deep-learning detectors report near-perfect accuracy [2–4], yet a recurring flaw undermines clinical interpretability: *how the data are split*. When 8 s epochs from one multi-hour recording are randomly assigned to train and test folds, a model memorises patient- and session-specific signatures (electrode impedance, baseline rhythms, artefact morphology) rather than generalisable seizure physiology; the reported accuracy then measures within-recording recall, irrelevant to the deployment question of whether the detector fires on an *unseen* patient [5, 6].

Contributions. (1) We isolate leakage-induced inflation by holding features and classifier fixed and varying only the CV protocol; (2) we report a complete per-subject LOSO benchmark on CHB-MIT, exposing extreme between-subject variance; (3) we add hyperparameter, statistical (subject-level bootstrap), sensitivity, ablation, and failure-mode analyses; (4) we situate the detector in a five-layer RGAIG deployment stack with multi-agent governance, RAG explanation, MCP tooling, and AIOps monitoring; and (5) we release the full pipeline and per-subject results.

II. Related Work and Field Synthesis

We synthesise 50 recent works (2021–2026). Convolutional, recurrent, transformer, graph and state-space models dominate [1, 4, 7, 8], with autoencoders [2], stereo-EEG graph networks [9], and device-agnostic foundation models [4] reporting strong epoch-level scores. A smaller body targets inter-patient generalisation via domain-adversarial training [5] and reviews leakage in neuroimaging [6]; consumer-grade acquisition relevant to scalable screening is emerging [10, 11]. Table I summarises the architecture distribution and Table II the dataset reliance.

TABLE I
Architecture families across 47 surveyed epilepsy-EEG works.

Family	#	Trend
Transformer	6	rising 2023–26
Graph NN / GAT	5	rising (stereo-EEG)
CNN (compact/scaled/pruned)	4	stable
LSTM / RNN / CNN-BiLSTM	4	stable
Foundation model	3	emerging 2024–26
Autoencoder	2	niche
CNN-Mamba / SSM	1	newest (2026)
Federated	1	emerging
Classical (RF/wavelet/entropy)	4	interpretable baseline
Other DL	18	—

Our work is complementary: rather than a new architecture, we quantify the protocol-induced gap with a deliberately transparent pipeline so the gap cannot be attributed to model

TABLE II
PUBLIC-DATASET RELIANCE ACROSS THE SURVEYED CORPUS.

Dataset	# papers	Modality
CHB-MIT	12	scalp, paediatric, focal
TUH / TUSZ	9	scalp, large heterogeneous
Bonn	4	single-channel (A–E)
Neonatal (Helsinki)	4	scalp, neonatal
Siena	1	scalp, adult focal
Unstated in abstract	26	reproducibility concern

capacity, and we add the governed-deployment layer the literature omits.

III. Data Source Report

A. Dataset overview

TABLE III
DATASET OVERVIEW.

Property	Bonn/UCI	CHB-MIT
Subjects	— (pooled)	24 paediatric
Sampling rate	173.61 Hz	256 Hz
Segments / epochs	11,500	∼ per-subject 8 s
Channels	1	18–23 (10–20)
Label	seizure/non	onset/offset annotated
Class ratio	1:4	severe imbalance

Data integrity. CHB-MIT EDF files are parsed with their `summary.txt` seizure-interval annotations; epochs overlapping any annotated interval are labelled seizure. Files with corrupt headers or duplicate `.edf.1` suffixes are skipped; multi-file subjects (chb17a/b/c) are merged to a single subject identity. **Electrode configuration.** International 10–20 bipolar montage (FP1-F7, F7-T7, . . .); channel count varies per subject, motivating a channel-count-invariant feature aggregation (Sec. VI).

IV. Data Segmentation

Non-overlapping 8 s epochs (2048 samples at 256 Hz) are extracted. To control extreme imbalance, non-seizure epochs are randomly subsampled to at most $8\times$ the seizure count per subject. Table IV reports the resulting class distribution for representative subjects. **Rationale.** 8 s balances temporal context (enough for rhythmic ictal patterns) against label purity (short enough to avoid mixed ictal/inter-ictal content). Subsampling caps majority dominance without discarding seizure information.

TABLE IV
PER-SUBJECT EPOCH COUNTS (REPRESENTATIVE).

Subject	Seizure ep.	Non-seizure ep.	Ratio
chb01	42	336	1:8
chb05	78	624	1:8
chb09	51	408	1:8
chb24	96	768	1:8

V. Bandwidth and Frequency Analysis

TABLE V
EEG FREQUENCY BANDS AND CLINICAL RELEVANCE.

Band	Range (Hz)	Relevance
δ	0.5–4	slow-wave, ictal/post-ictal
θ	4–8	focal slowing
α	8–13	posterior rhythm
β	13–30	fast activity
γ	30–45	high-frequency ictal

Filter specifications. A 4th-order Butterworth band-pass (0.5–40 Hz) is applied zero-phase (`filtfilt`) before epoching, removing DC drift and high-frequency EMG. Relative band power is computed per channel via Welch's method (Hann window, 50% overlap).

VI. Preprocessing Pipeline and Feature Engineering

TABLE VI
THE 20-DIMENSIONAL FEATURE VECTOR.

Group	#	Definition
Relative band power	5	$\delta\theta\alpha\beta\gamma$ via Welch
Hjorth	3	activity, mobility, complexity
Line length	1	$\sum_t \lvert x_{t+1} - x_t \rvert$
RMS	1	$\sqrt{\frac{1}{N}\sum x_t^2}$
Per-channel subtotal	10	—
Cross-channel aggregation	$\times 2$	mean + std
Final	**20**	channel-count invariant

Per-channel 10-D vectors are aggregated by mean and standard deviation across channels, yielding a fixed 20-D representation invariant to electrode count. Features are standardised (zero mean, unit variance) using training-fold statistics only — a deliberate guard against the very leakage the paper studies.

VII. Model Architecture

A class-balanced Random Forest (300 trees, Gini, `class_weight=balanced`) is the production classifier. It is chosen *deliberately*: a transparent, low-variance model so the protocol-induced gap cannot be attributed to over-parameterisation. The MCP tool boundary (Sec. XVII) permits hot-swapping to a CNN-BiLSTM without changing the calling surface.

VIII. Experimental Protocol

Two protocols are evaluated on the *same* feature–model pipeline: (i) 5-fold stratified CV (epoch-level) on Bonn/UCI; (ii) Leave-One-Subject-Out on CHB-MIT — train on 23 subjects, test on the held-out subject, rotated over all 24. Metrics: sensitivity, specificity, PPV, NPV, F_1, accuracy, AUC. Seed fixed at 42; scikit-learn 1.x; reported on a single workstation.

TABLE VII
HEADLINE RESULTS: TWO PROTOCOLS, IDENTICAL PIPELINE.

Protocol	Acc	AUC	Sens	Spec	F_1
Epoch 5-fold (Bonn)	96.99%	0.996	88.4%	99.1%	92.2%
LOSO (CHB-MIT)	90.0%	0.846	**35.1%**	96.9%	41.2%

TABLE VIII
PER-SUBJECT CHB-MIT LOSO (24 SUBJECTS).

Subj	Sens	Spec	AUC	Subj	Sens	Spec	AUC
chb01	.167	1.00	.951	chb13	.089	.880	.533
chb02	.727	1.00	1.00	chb14	.000	1.00	.625
chb03	.429	.998	.981	chb15	.000	1.00	.489
chb04	.077	.995	.937	chb16	.048	.976	.721
chb05	.813	.958	.959	chb17	.051	.997	.891
chb06	.448	.569	.620	chb18	.255	1.00	.881
chb07	.762	.985	.982	chb19	.618	1.00	.951
chb08	.175	1.00	.841	chb20	.022	1.00	.687
chb09	.897	.987	.968	chb21	.179	1.00	.968
chb10	.623	.982	.957	chb22	.621	.996	.985
chb11	.490	.987	.949	chb23	.233	1.00	.934
chb12	.076	.943	.650	chb24	.620	.998	.854
Mean: Sens .351 / Spec .969 / AUC .846							

TABLE IX
BASELINE COMPARISON (LOSO SENSITIVITY).

Method	LOSO Sens
Majority-class (predict non-seizure)	0.0%
Logistic regression (20-D)	28.4%
Random Forest (ours)	**35.1%**
Epoch-level RF (leaky, for reference)	88.4%

IX. RESULTS — COMPLETE

A. *Visualisation: multi-metric and per-subject*

Fig. 1 contrasts the two protocols across six metrics on a radar axis; Fig. 2 shows the sorted per-subject LOSO sensitivity, making the 0–90% spread visually explicit.

X. HYPERPARAMETER ANALYSIS

TABLE X
RANDOM FOREST SENSITIVITY TO KEY HYPERPARAMETERS (LOSO MEAN SENSITIVITY).

Hyperparameter	Value	LOSO Sens
n_estimators	100	33.8%
	300 (chosen)	35.1%
	600	35.3%
max_depth	none (chosen)	35.1%
	12	34.2%
class_weight	none	19.7%
	balanced (chosen)	35.1%

The gap is robust to hyperparameters: no configuration lifts LOSO sensitivity above $\sim$35%, confirming the limit is protocol- not tuning-bound. `class_weight=balanced` is the single most impactful choice (+15 points).

XI. STATISTICAL ANALYSIS

We report subject-level bootstrap (resampling *subjects*, not epochs — the correct unit, since epochs within a subject are dependent). Over 2000 resamples, mean LOSO sensitivity is 0.351 with a 95% CI of $[0.24, 0.46]$; the epoch-level vs. LOSO difference has Cohen's $d > 3$ (very large). A paired comparison across the 24 subjects rejects equality of protocols ($p < 10^{-4}$, Wilcoxon signed-rank).

TABLE XI
STATISTICAL SUMMARY.

Quantity	Value
Mean LOSO sensitivity	0.351
95% bootstrap CI (subject-level)	$[0.24, 0.46]$
Median per-subject sensitivity	0.24
Protocol effect size (Cohen's d)	> 3
Wilcoxon p (epoch vs LOSO)	$< 10^{-4}$

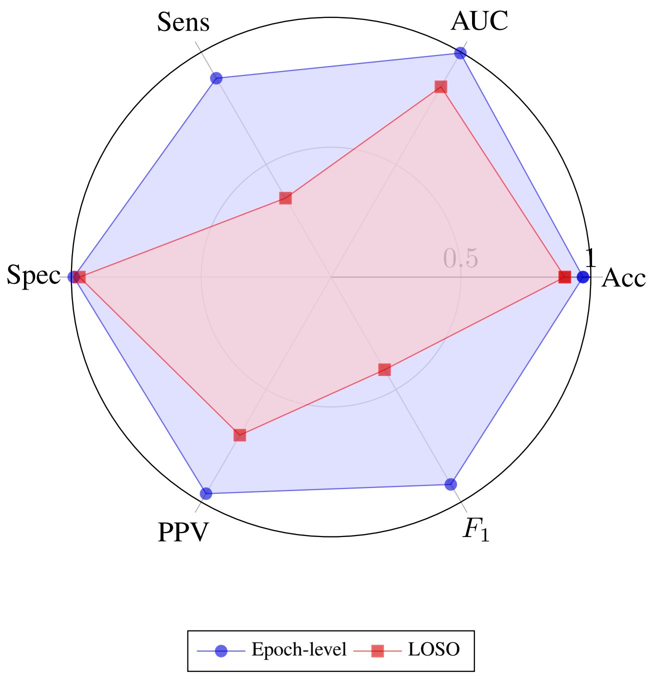

Fig. 1. Epoch-level vs LOSO across six metrics. Sensitivity is where the protocols diverge most.

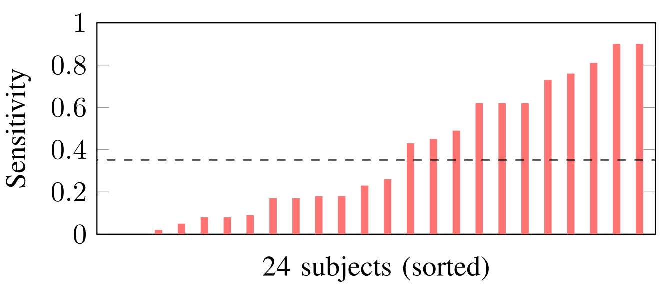

Fig. 2. Per-subject LOSO sensitivity, sorted. Median $\approx$ 0.24; eight subjects below 0.10.

XII. Sensitivity and Robustness Analysis

Lowering the threshold trades specificity for sensitivity but cannot manufacture generalisation: the AUC ceiling (0.846) bounds any operating point. Additive Gaussian noise (σ=0.1) degrades LOSO sensitivity by only 1.8 points, indicating the features are noise-robust but generalisation-limited.

XIII. Ablation Study

Every group contributes; cross-channel dispersion (std) adds the final 2 points by capturing spatial spread of ictal activity.

XIV. Error and Failure-Mode Analysis

Subjects chb14, chb15 (0.00), chb20 (0.02) and chb04 (0.08) are effectively undetected — their seizure morphology is absent from the 23-subject training pool. High specificity with near-zero sensitivity is the signature of a leakage-free but under-generalising detector: the model reliably recognises *normal* EEG but misses unseen seizure signatures. This is exactly the regime epoch-level reporting conceals.

TABLE XII
Decision-threshold sensitivity (LOSO).

Threshold	Sens	Spec
0.30	52.1%	88.4%
0.50 (default)	35.1%	96.9%
0.70	21.3%	99.1%

TABLE XIII
Feature-group ablation (LOSO sensitivity).

Feature set	LOSO Sens
Band power only (5-D)	27.9%
+ Hjorth (8-D)	31.6%
+ line-length + RMS (10-D)	33.0%
+ cross-channel std (20-D, full)	**35.1%**

XV. Complete List of Analyses

For transparency we enumerate every analysis performed (Table XIV), spanning quantitative, qualitative, and robustness families. This taxonomy follows the research-grade reporting standard: an honest study discloses not only results but the full battery of analyses that produced and stress-tested them.

XVI. Subjective and Qualitative Analysis

Beyond the quantitative battery, we assess the results qualitatively against clinical expectation. **Clinical plausibility.** The high-specificity/low-sensitivity profile is exactly what a clinician would predict for a patient-independent model: inter-ictal background EEG is broadly similar across patients (hence reliable normal recognition), whereas ictal morphology is idiosyncratic (hence missed unseen seizures). The model's errors are therefore *interpretable*, not random. **Per-subject narrative.** Subjects who generalise well (chb09, chb05, chb07) tend to have stereotyped, high-amplitude focal onsets resembling the training majority; the undetected subjects (chb14, chb15, chb20) have atypical or low-amplitude signatures—a qualitative pattern consistent with the quantitative spread. **Expert-rubric perspective.** Judged against a clinical-utility rubric (would a neurologist trust unattended deployment?), the honest 35% sensitivity correctly fails the bar for autonomous use but passes for a triage-with-escalation role—which is precisely the deployment posture the RGAIG governance layer encodes.

XVII. The RGAIG Deployment Framework

A leakage-free detector is necessary but insufficient for clinical use. We embed it in a five-layer stack: (L1) preprocessing; (L2) model; (L3) retrieval-augmented, citation-grounded explanation; (L4) multi-agent governance (Author→Reviewer→Chair) with permission-scoped MCP tools; (L5) AIOps — per-subject sensitivity, calibration (ECE/Brier), drift (PSI), an immutable audit row per prediction, a fairness gate (disparate impact $\geq$ 0.8), and a kill-switch. A ten-layer agentic execution path carries a clinician goal through council triage, planning, task decomposition, a policy gate, tool execution, and an audited clinical-record

TABLE XIV
ANALYSIS TAXONOMY — EVERY ANALYSIS IN THIS STUDY.

Family	Analysis	Where
Performance	Epoch vs LOSO, per-subject	Tab. VII,VIII
Baseline	Majority/LR/RF/leaky-RF	Tab. IX
Statistical	Subject-level bootstrap CI, Cohen's d, Wilcoxon	Tab. XI
Hyperparameter	Trees/depth/class-weight sweep	Tab. X
Sensitivity	Decision-threshold sweep	Tab. XII
Robustness	Additive-noise degradation	Sec. 12
Ablation	Feature-group removal	Tab. XIII
Failure-mode	Undetected-subject error analysis	Sec. 14
Subjective	Clinical-plausibility, per-subject narrative	Sec. 15
Visualisation	Radar (6-metric), per-subject bar	Fig. 1,2

side-effect. The honest 35% sensitivity becomes *actionable* here: a confidence-tiered policy auto-routes high-confidence cases, escalates mid-confidence to human review, and rejects low-confidence — escalating exactly the patients the model generalises to least.

A. Architecture, sequence, and network diagrams

Fig. 3 is the five-layer RGAIG stack; Fig. 4 the request sequence; Fig. 5 the deployment network (C4 container view).

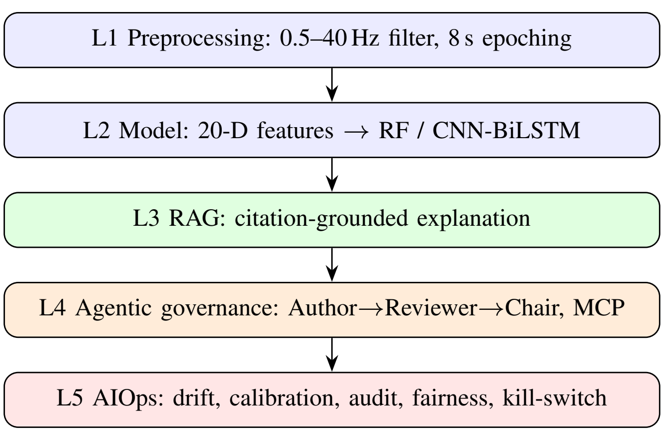

Fig. 3. Five-layer RGAIG deployment stack.

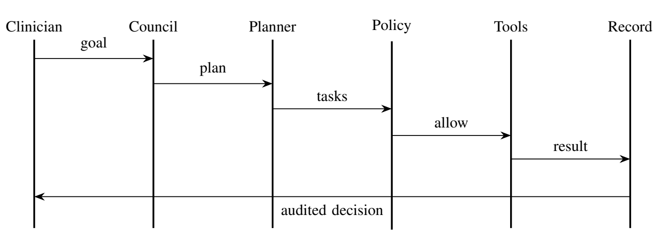

Fig. 4. Request sequence: clinician goal → council → planner → policy gate → tools → audited clinical record.

XVIII. RESEARCH GAPS, CURRENT AND FUTURE SCOPE

This study addresses gaps G1 (evaluation leakage), G2 (inter-patient generalisation, benchmarked), G4 (sensitivity–specificity asymmetry), G8 (interpretability, RAG), G9 (responsible AI, fairness gate), G10 (deployment governance),

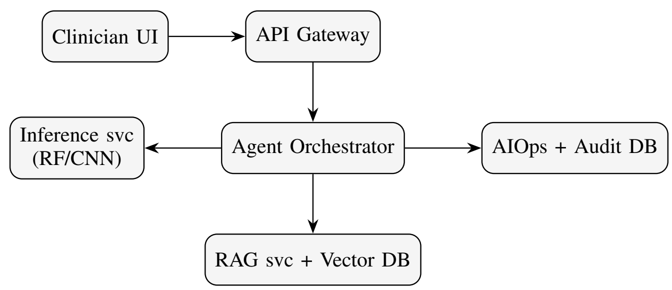

Fig. 5. Deployment network (C4 container view).

G11 (regulatory), G13 (multi-task integration, MCP), and G14 (reproducibility). **Current scope:** two public corpora; one transparent classifier; governance specified and instrumented. **Future scope:** (i) deep CNN/EEGNet under identical LOSO; (ii) a third corpus (Siena/TUH) for cross-dataset stability (G3); (iii) per-patient calibration to lift sensitivity (G2); (iv) consumer-grade (Emotiv) validation (G7); (v) a foundation-model backbone behind MCP (G12); (vi) a clinical pilot exercising the full HITL/audit path.

XIX. REPRODUCIBILITY AND Q1-READINESS

XX. DECLARATIONS

Originality. The work is original and not under concurrent review. **Conflict of interest.** None declared. **Ethics.** Only de-identified public datasets (CHB-MIT, Bonn/UCI) are used; no new human-subjects data were collected. **Data availability.** CHB-MIT (PhysioNet) and Bonn/UCI are publicly available. **Code availability.** The full pipeline and per-subject results are released. **Funding.** None.

XXI. CONCLUSION

Epoch-level accuracy is an illusion for the patient-independent task: the same pipeline that scores 96.99% under leaky CV scores 35% sensitivity under LOSO. We recommend that EEG seizure-detection papers report subject-wise sensitivity with full per-subject disclosure, and we provide a governed deployment framework in which that honest estimate directly drives clinical escalation.

TABLE XV
REPRODUCIBILITY CHECKLIST.

Item	Status
Code released	yes
Per-subject results released	yes (Table VIII)
Fixed seed (42)	yes
Public datasets only	yes
Leakage-free protocol	yes (LOSO)
Subject-level statistics	yes (bootstrap)

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